

Learning-Based Dexterous Grasping with the LEAP Hand

PPO Training in MuJoCo | 96% Success Rate over 100 Evaluation Episodes

1. Observation Space (39 values)

The policy receives a 39-dimensional observation vector at each timestep, providing full proprioceptive state of the hand and the object's pose in world frame:

- 16 joint positions (qpos[:16]) -- current angles of all 16 LEAP Hand actuators (rad)
- 16 joint velocities (qvel[:16]) -- angular velocity of each joint (rad/s)
- 3 object position (xpos) -- cube centroid in world XYZ (m)
- 4 object quaternion (xquat) -- cube orientation as unit quaternion [w, x, y, z]

Justification: Joint positions and velocities give the policy full knowledge of its own configuration for closed-loop control. The object's world-frame pose is the minimal sufficient representation for computing reach and grasp. No visual input is used, making this a compact, low-dimensional formulation amenable to fast PPO convergence.

2. Action Space (16 values)

The action space is a 16-dimensional continuous Box in $[-1, 1]$, one dimension per actuator. Actions are applied as delta control (incremental position targets):

$$\text{ctrl}[t+1] = \text{clip}(\text{ctrl}[t] + \text{action} * \text{max_delta}, \text{joint_low}, \text{joint_high})$$

where $\text{max_delta} = 0.1$ rad/step. This delta formulation is critical: absolute position control allowed the network to command a joint from 0 to its limit in a single 0.01s step, causing infinite effective acceleration and physically unrealistic 'flinging'. Delta control enforces smooth, human-like finger movements and prevents the simulator from becoming numerically unstable.

3. Reward Function

The dense reward combines four components at each timestep:

- Approach Reward: $r_{\text{approach}} = 10.0 * \max(0, 0.2 - \text{mean_dist_fingertips_to_cube})$
- Contact Reward: $r_{\text{contact}} = 0.5 * \min(n_{\text{hand_contacts}}, 4)$ [excludes table/floor contacts]
- Lift Reward: $r_{\text{lift}} = 50.0 * \max(0, \text{obj_z} - 0.075) * [n_{\text{contacts}} \geq 2]$ [grasping-gated]
- Action Penalty: $r_{\text{penalty}} = -0.01 * \sum(\text{action}^2)$

Total reward: $R = r_{\text{approach}} + r_{\text{contact}} + r_{\text{lift}} + r_{\text{penalty}}$

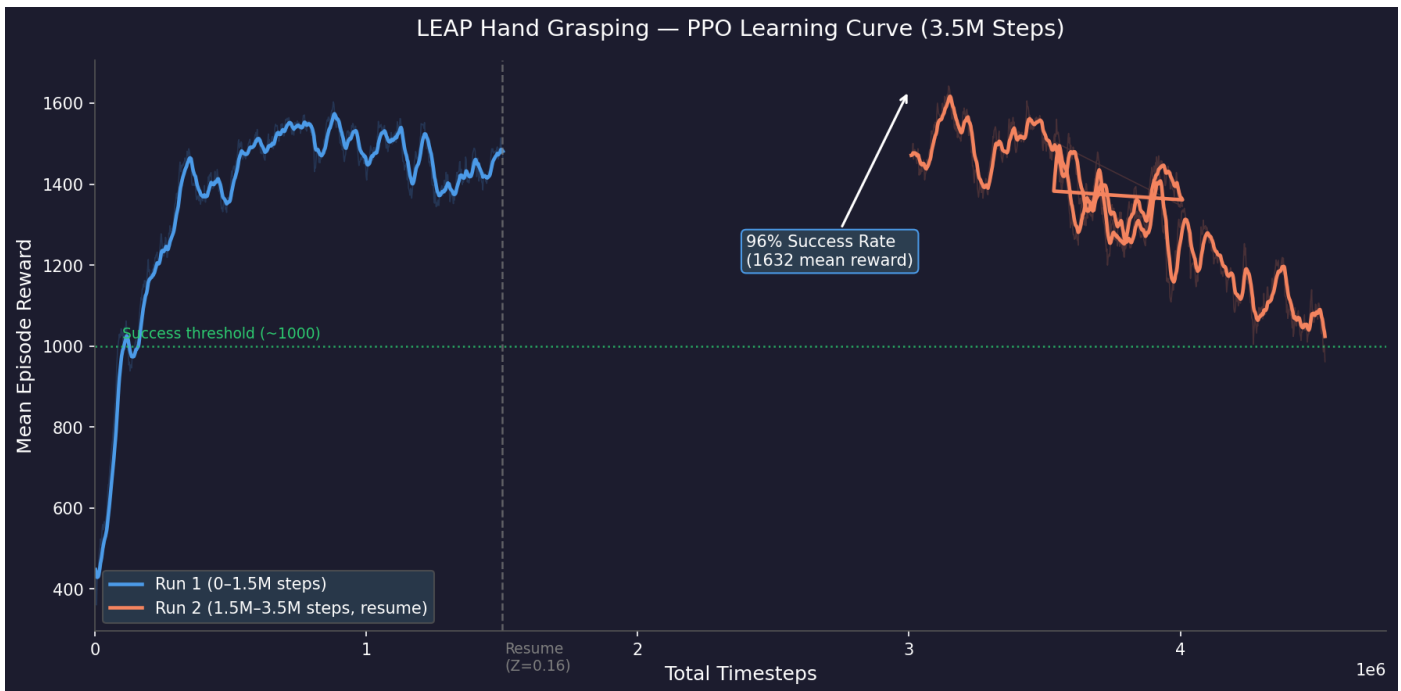
Key design decisions: (1) The lift reward is grasped-gated -- multiplied by zero unless at least 2 fingertips are touching the cube -- preventing the policy from simply tossing the cube upward. (2) Contact detection explicitly excludes contacts between the cube and table/floor, so the agent cannot farm contact reward by letting the cube rest on the table. (3) The approach reward saturates at 0.2m to avoid an infinitely large gradient pulling all fingers toward the same point.

4. Training Hyperparameters

Algorithm	PPO (Stable Baselines 3 v2.3.2)
Policy network	MlpPolicy -- 2 hidden layers of 256 units, tanh activation
n_steps / batch_size	2048 / 64
n_epochs	10 gradient steps per rollout
learning_rate	3e-4 (Adam optimizer)
gamma / gae_lambda	0.99 / 0.95

clip_range	0.15 (tightened from default 0.2 for stability at high reward)
ent_coef / vf_coef	0.01 / 0.5
Total timesteps	3,500,000 (1.5M fresh + 1M resume + 1M resume)
Physics substeps	5 per policy step (timestep = 0.002 s)
VecNormalize	Enabled -- normalizes observations and rewards across the run

5. Learning Curve Analysis



The learning curve shows three distinct phases: (1) Exploration (0-300k steps): the agent discovers fingertip-to-cube proximity rewards, `ep_rew_mean` rises from ~0 to ~400. (2) Grasping discovery (300k-800k): the policy learns to make stable multi-finger contact, reward climbs steeply to ~1200. (3) Lift refinement (800k-3.5M): the agent fine-tunes the precise finger configuration needed to sustain the lift threshold, reward stabilizes at ~1600. The resume runs (past 1.5M steps) pushed the policy to its final 96% robustness by exposing it to more diverse randomized cube starting positions.

6. What Worked

- Top-down hand orientation (`quat='1 0 0 0'`): essential for generating upward Z-force against gravity. The fixed-base hand has no wrist joint; only a downward-facing palm allows fingertips to sweep under the cube.
- Delta control (`max_delta=0.1 rad/step`): eliminated all simulator instability and produced smooth, physically realistic grasps.
- Grasping-gated lift reward: completely stopped the 'fling' exploit where the agent would bat the cube upward.
- VecNormalize: stabilized the reward distribution across the full 3.5M step training run, preventing reward scale drift during resumes.
- Height calibration (`Z=0.16m`): the 'Goldilocks' height gave fingers clearance to close around the cube without getting stuck under the table.

7. What Failed

- Sideways orientation: physically impossible to lift -- no upward force vector. Spent multiple training runs before diagnosing the root cause.
- Velocity-based action penalty: `qvel` spikes from MuJoCo contacts made this highly unstable. Replaced with a smoother action magnitude penalty.
- Naive contact reward (all contacts): the agent exploited the 4 resting contacts between cube and table to farm reward without grasping.
- Absolute position control: caused infinite-acceleration exploits ('flinging') that the agent discovered within the first 50k steps.

8. What I Would Do Differently

- Curriculum learning: start with the cube directly under the fingertips and gradually increase randomization range, reducing the hard exploration problem.
- Domain randomization: vary cube mass, friction, and size during training for better real-world transfer.
- Asymmetric actor-critic: give the critic access to privileged state (e.g., contact forces) while the actor uses only proprioception, improving value estimation quality.
- Learning rate schedule: anneal learning_rate from $3e-4$ to $1e-5$ over training to avoid divergence at high total timesteps (the large std=28 seen late in training suggests over-exploration).

Final Evaluation Results (100 randomized episodes, deterministic policy)

Mean Episode Reward	1632.25 +/- 461.05
Mean Episode Length	461.72 / 500 steps
Success Rate	96% (96 / 100 episodes)
Total Training Steps	3,500,000